COMP0086 Probabilistic and Unsupervised Learning
2021–22 Final Examination — Solutions

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1 Conjugacy in Regression

1.1 Conjugate Prior

Definition 1.1 (Conjugate prior)**.** A family of priors Π is **conjugate** to a likelihood family $\{p(\mathcal{D} \mid \theta)\}$ if for every prior $\pi \in \Pi$ the posterior $p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)\, \pi(\theta)$ also lies in Π.

The practical benefit is that Bayesian updating reduces to a closed-form mapping of prior hyper-parameters to posterior hyper-parameters.

1.2 Conjugate Prior on $\mathbf{w}$ when σ^2 is Known

The likelihood $p(y_n \mid \mathbf{x}_n, \mathbf{w}, \sigma^2) = \mathcal{N}(y_n; \mathbf{w}^\top \mathbf{x}_n, \sigma^2)$ is Gaussian and quadratic in $\mathbf{w}$. The conjugate prior is therefore Gaussian. Following the parameterisation used in the lecture notes (Bayesian linear regression),

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \mathbf{0}, \mathbf{A}^{-1}), \tag{1}$$

with prior precision matrix $\mathbf{A}$ and zero prior mean.

1.3 Joint Posterior under Independent Priors

Stack the data as $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N] \in \mathbb{R}^{D\times N}$ and $Y = [y_1, \dots, y_N] \in \mathbb{R}^{1\times N}$. The Gaussian likelihood is

$$p(Y \mid \mathbf{X}, \mathbf{w}, \sigma^2) = (2\pi\sigma^2)^{-N/2} \exp\Big[-\tfrac{1}{2\sigma^2}\, (Y - \mathbf{w}^\top \mathbf{X})(Y - \mathbf{w}^\top \mathbf{X})^\top\Big].$$

With independent priors $p(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \mathbf{0}, \mathbf{A}^{-1})$ and $p(\sigma^2) = \mathrm{InvGamma}(\sigma^2; \alpha, \beta)$, the unnormalised joint posterior is

$$\begin{aligned} p(\mathbf{w}, \sigma^2 \mid \mathcal{D}) \propto\ & (\sigma^2)^{-\alpha-1} \exp\Big[-\tfrac{\beta}{\sigma^2}\Big] \\ & \times (\sigma^2)^{-N/2} \exp\Big[-\tfrac{1}{2\sigma^2}(Y - \mathbf{w}^\top \mathbf{X})(Y - \mathbf{w}^\top \mathbf{X})^\top\Big] \exp\Big[-\tfrac{1}{2}\mathbf{w}^\top \mathbf{A}\mathbf{w}\Big]. \end{aligned}$$

Discarding constants and isolating the dependence on $(\mathbf{w}, \sigma^2)$,

$$p(\mathbf{w}, \sigma^2 \mid \mathcal{D}) \ \propto\ (\sigma^2)^{-\alpha-1-N/2} \exp\Big[-\tfrac{\beta}{\sigma^2} - \tfrac{1}{2\sigma^2}\|Y - \mathbf{w}^\top \mathbf{X}\|^2 - \tfrac{1}{2}\mathbf{w}^\top \mathbf{A}\mathbf{w}\Big]. \tag{2}$$

Remark 1.2 (Why this prior is not conjugate). Conditional on σ^2, $p(\mathbf{w} \mid \sigma^2, \mathcal{D})$ is Gaussian with precision $\mathbf{A} + \mathbf{X}\mathbf{X}^\top/\sigma^2$. This precision mixes a σ^2-independent term ($\mathbf{A}$) with a $1/\sigma^2$ term, so the conditional cannot be written as $\mathcal{N}(\mathbf{w} \mid \boldsymbol{\mu}', \sigma^2(\mathbf{A}')^{-1})$ for any single $\mathbf{A}'$. Hence the joint posterior is not in the Normal-Inverse-Gamma family, and the independent prior is not conjugate.

1.4 Conjugacy of the Joint Normal-Inverse-Gamma Prior

Take $\mu = \mathbf{0}$. The prior is

$$p(\mathbf{w}, \sigma^2) = \mathrm{InvGamma}(\sigma^2; \alpha, \beta)\, \mathcal{N}(\mathbf{w} \mid \mathbf{0}, \sigma^2 \mathbf{A}^{-1}).$$

The Gaussian normaliser contributes a factor $(\sigma^2)^{-D/2} |\mathbf{A}|^{1/2}$, so multiplying with the likelihood:

$$p(\mathbf{w}, \sigma^2 \mid \mathcal{D}) \propto (\sigma^2)^{-\alpha-1-D/2-N/2} \exp\Big[-\tfrac{1}{\sigma^2}\Big(\beta + \tfrac{1}{2}\mathbf{w}^\top \mathbf{A}\mathbf{w} + \tfrac{1}{2}(Y - \mathbf{w}^\top \mathbf{X})(Y - \mathbf{w}^\top \mathbf{X})^\top\Big)\Big].$$

Crucially, the entire $\mathbf{w}$-quadratic is now scaled by $1/\sigma^2$. Completing the square in $\mathbf{w}$:

$$\begin{aligned} \mathbf{w}^\top \mathbf{A}\mathbf{w} + (Y - \mathbf{w}^\top \mathbf{X})(Y - \mathbf{w}^\top \mathbf{X})^\top &= \mathbf{w}^\top(\mathbf{A} + \mathbf{X}\mathbf{X}^\top)\mathbf{w} - 2\mathbf{w}^\top \mathbf{X} Y^\top + Y Y^\top \\ &= (\mathbf{w} - \mu')^\top \mathbf{A}'(\mathbf{w} - \mu') + Y Y^\top - \mu'^\top \mathbf{A}' \mu', \end{aligned}$$

with

$$\mathbf{A}' = \mathbf{A} + \mathbf{X}\mathbf{X}^\top, \qquad \mu' = \mathbf{A}'^{-1} \mathbf{X} Y^\top. \tag{3}$$

Substituting back, the joint posterior factorises as

$$p(\mathbf{w}, \sigma^2 \mid \mathcal{D}) = \mathrm{InvGamma}(\sigma^2; \alpha', \beta')\, \mathcal{N}(\mathbf{w} \mid \mu', \sigma^2 \mathbf{A}'^{-1}),$$

which is in the same Normal-Inverse-Gamma family as the prior. Reading off the hyper-parameters:

$$\boxed{\mathbf{A}' = \mathbf{A} + \mathbf{X}\mathbf{X}^\top, \qquad \mu' = \mathbf{A}'^{-1}\mathbf{X}Y^\top, \qquad \alpha' = \alpha + \tfrac{N}{2}, \qquad \beta' = \beta + \tfrac{1}{2}\big(YY^\top - \mu'^\top \mathbf{A}'\mu'\big).} \tag{4}$$

Hence the joint Normal-Inverse-Gamma is conjugate.

1.5 Type-II ML over A: Effect of Marginalising σ^2

In lecture, hyper-parameters in $\mathbf{A}$ were learned by maximising the marginal likelihood

$$p(\mathcal{D} \mid \mathbf{A}, \sigma^2) = \int p(\mathcal{D} \mid \mathbf{w}, \sigma^2)\, p(\mathbf{w} \mid \mathbf{A})\, d\mathbf{w},$$

jointly optimised over σ^2. If we instead integrate out *both* $\mathbf{w}$ and σ^2 (using the conjugate joint prior), the evidence becomes

$$p(\mathcal{D} \mid \mathbf{A}) = \iint p(\mathcal{D} \mid \mathbf{w}, \sigma^2)\, p(\mathbf{w}, \sigma^2 \mid \mathbf{A})\, d\mathbf{w}\, d\sigma^2,$$

a Student-t-like quantity proportional to $|\mathbf{A}|^{1/2} |\mathbf{A}'|^{-1/2}\, (\beta')^{-\alpha'} \Gamma(\alpha')$ with $\mathbf{A}', \beta'$ from (4). The dependence on $\mathbf{A}$ enters through both $|\mathbf{A}|^{1/2}|\mathbf{A}'|^{-1/2}$ *and* the data-dependent factor $\beta'(\mathbf{A})$, whereas the joint maximisation in lecture only profiles over σ^2.

I therefore expect the optima to be **different in general**: marginalising σ^2 averages over its posterior uncertainty, while joint maximisation plugs in a point estimate $\hat{\sigma}^2(\mathbf{A})$. The two coincide only in the limit where the σ^2 posterior is tightly concentrated (large N, sharp evidence), in which case the Laplace approximation to the σ^2 integral matches the profile likelihood.

2 Optimisation

2.1 Decoupling Lemma

Lemma 2.1 (Separability)**.** *If $f_k : \mathbb{R}^d \to \mathbb{R}$ are convex and $g_k : \mathbb{R}^d \to \mathbb{R}$ are linear, problem (A) and the K separate problems (B) share the same optimal solutions.*

Proof. The objective $\sum_k f_k(x_k)$ separates additively in disjoint variable blocks x_k, and the constraint set is the Cartesian product $\prod_k \{x_k : g_k(x_k) = 0\}$ since each g_k involves only x_k. Minimisation over a product set of a separable objective is achieved variable-block by variable-block:

$$\min_{x_1,\ldots,x_K} \sum_k f_k(x_k) \;=\; \sum_k \min_{x_k : g_k(x_k)=0} f_k(x_k).$$

Convexity of each f_k together with linearity (hence convexity) of g_k ensures each sub-problem is convex, and stacking sub-problem minimisers yields a global minimiser of (A). $\square$

2.2 MLE as a Convex Programme

The log-likelihood under the model M (with P_{ij} the column-stochastic transition matrix) is

$$\log p(x_1, \ldots, x_n \mid P) = \log p(x_1) + \sum_{i,j} N_{ij} \log P_{ij}. \tag{5}$$

With the initial distribution unconstrained, the term $\log p(x_1)$ does not couple with P and is dropped. Maximising (5) in P is equivalent to:

$$\min_{P \geq 0} \; -\sum_{i,j} N_{ij} \log P_{ij} \quad \text{subject to} \quad \sum_{i \leq K} P_{ij} = 1 \;\forall j. \tag{6}$$

The objective is a non-negative combination of $-\log$ (each convex), hence convex; the equality constraints are affine. So (6) is a convex programme whose solution is $\hat{P}$.

2.3 Per-Column Solution

For a single column j, the relevant sub-problem is

$$\min_{P_{1j},\ldots,P_{Kj} \geq 0} \; -\sum_{k \leq K} N_{kj} \log P_{kj} \quad \text{subject to} \quad \sum_{k \leq K} P_{kj} = 1. \tag{7}$$

Writing the Lagrangian $L(P_{\cdot j}, \lambda) = -\sum_k N_{kj} \log P_{kj} + \lambda(\sum_k P_{kj} - 1)$ and setting $\partial L / \partial P_{kj} = 0$:

$$-\frac{N_{kj}}{P_{kj}} + \lambda = 0 \implies P_{kj} = \frac{N_{kj}}{\lambda}.$$

Imposing $\sum_k P_{kj} = 1$ gives $\lambda = \sum_{i \leq K} N_{ij}$, hence

$$\hat{P}_{kj} = \frac{N_{kj}}{\sum_{i \leq K} N_{ij}}. \tag{8}$$

Since the objective in (7) is strictly convex on the simplex, this stationary point is the unique global minimiser. Lemma 2.1 applies to (6): the objective is a sum $\sum_j h_j(P_{\cdot j})$ of column-wise terms and the constraints involve only one column each, so the column-wise solutions (8) stack to give $\hat{P}$.

2.4 Adding the Invariance Constraint

Imposing additionally $\pi P = \pi$, the convex programme becomes

$$\min_{P\geq 0} -\sum_{i,j} N_{ij}\log P_{ij} \quad \text{s.t.} \quad \sum_i P_{ij} = 1\ \forall j, \qquad \sum_j P_{ij}\pi_j = \pi_i\ \forall i.$$

The objective is still convex and separable across columns, but the new constraint $\sum_j \pi_j P_{ij} = \pi_i$ couples *all* columns through the index j. Lemma 2.1 therefore **cannot** be used to split the problem into K column-wise sub-problems.

3 Derivatives in the EM Algorithm

3.1 Free Energy and the Lower Bound

Definition 3.1 (Free energy)**.** For a variational distribution $q(\mathcal{Z})$ over the latent variables,

$$\mathcal{F}(q,\theta) = \mathbb{E}_q[\log p(\mathcal{X},\mathcal{Z}\mid\theta)] + H[q] = \int q(\mathcal{Z})\log\frac{p(\mathcal{X},\mathcal{Z}\mid\theta)}{q(\mathcal{Z})}\,d\mathcal{Z}. \tag{9}$$

Using $\log p(\mathcal{X}\mid\theta) = \log p(\mathcal{X},\mathcal{Z}\mid\theta) - \log p(\mathcal{Z}\mid\mathcal{X},\theta)$ and integrating against q:

$$\ell(\theta) = \mathcal{F}(q,\theta) + \mathrm{KL}(q(\mathcal{Z})\,\|\,p(\mathcal{Z}\mid\mathcal{X},\theta)) \geq \mathcal{F}(q,\theta),$$

because $\mathrm{KL}\geq 0$ with equality iff $q(\mathcal{Z}) = p(\mathcal{Z}\mid\mathcal{X},\theta)$.

3.2 Equality of M-step and Likelihood Gradients

After an exact E-step, $q^\star(\mathcal{Z}) = p(\mathcal{Z}\mid\mathcal{X},\theta)$. Differentiating the free energy with $q^\star$ *held fixed* (i.e. at the current θ, regardless of how θ subsequently varies in the M-step):

$$\begin{aligned}
\nabla_\theta\mathcal{F}(q^\star,\theta) &= \nabla_\theta\int q^\star(\mathcal{Z})\log p(\mathcal{X},\mathcal{Z}\mid\theta)\,d\mathcal{Z} = \int q^\star(\mathcal{Z})\,\frac{\nabla_\theta p(\mathcal{X},\mathcal{Z}\mid\theta)}{p(\mathcal{X},\mathcal{Z}\mid\theta)}\,d\mathcal{Z}\\
&= \int \frac{p(\mathcal{X},\mathcal{Z}\mid\theta)}{p(\mathcal{X}\mid\theta)}\cdot\frac{\nabla_\theta p(\mathcal{X},\mathcal{Z}\mid\theta)}{p(\mathcal{X},\mathcal{Z}\mid\theta)}\,d\mathcal{Z} = \frac{1}{p(\mathcal{X}\mid\theta)}\int\nabla_\theta p(\mathcal{X},\mathcal{Z}\mid\theta)\,d\mathcal{Z}\\
&= \frac{\nabla_\theta\int p(\mathcal{X},\mathcal{Z}\mid\theta)\,d\mathcal{Z}}{p(\mathcal{X}\mid\theta)} = \frac{\nabla_\theta p(\mathcal{X}\mid\theta)}{p(\mathcal{X}\mid\theta)} = \nabla_\theta\ell(\theta).
\end{aligned}$$

Regularity is needed only to interchange differentiation and integration.

3.3 M-Step for the Mean of a Spherical Gaussian Mixture

Let $r_{nk} = p(z_n = k\mid\mathbf{x}_n,\theta)$ be the responsibilities. The terms of the expected complete-data log-likelihood depending on μ_k are

$$\mathcal{F}(\mu_k) = -\tfrac{1}{2}\sum_n r_{nk}\|\mathbf{x}_n-\mu_k\|^2 + \text{const}.$$

Gradient:

$$\nabla_{\mu_k}\mathcal{F} = \sum_n r_{nk}(\mathbf{x}_n-\mu_k). \tag{10}$$

Setting $\nabla_{\mu_k}\mathcal{F} = 0$ gives the exact M-step:

$$\mu_k^{\text{new}} = \frac{\sum_n r_{nk}\mathbf{x}_n}{\sum_n r_{nk}}. \tag{11}$$

The change effected by the M-step is

$$\mu_k^{\text{new}} - \mu_k^{\text{old}} = \frac{\sum_n r_{nk}(\mathbf{x}_n - \mu_k^{\text{old}})}{\sum_n r_{nk}} = \frac{1}{\sum_n r_{nk}} \nabla_{\mu_k} \mathcal{F}\Big|_{\mu_k^{\text{old}}},$$

i.e. a positive scalar multiple of the gradient. Hence the M-step moves μ_k along the gradient direction.

3.4 Is This Generally True?

No. The exact M-step jumps to the global maximiser of $\mathcal{F}$ in θ, not necessarily along the gradient at the starting point. Alignment with the gradient holds only when $\mathcal{F}$, viewed as a function of θ, is a quadratic with Hessian proportional to the identity, as for a spherical Gaussian mean. For other parameters (covariance matrices, mixing proportions, HMM transition rows, etc.) the exact M-step is equivalent to a Newton step pre-multiplied by an inverse Hessian that is not proportional to the identity, which generally rotates the update away from the steepest-ascent direction.

4 Markov Chains and Sampling Algorithms

4.1 When is $(G_n(x))_n$ a Markov Chain?

The recursion $G_{n+1}(x) = f_{\Theta_{n+1}}(G_n(x))$ shows G_{n+1} depends on the past only through G_n *and* on the new parameter Θ_{n+1}. Hence (G_n) is Markov iff Θ_{n+1} is independent of $G_1, \dots, G_n$ given G_n.

- **i.i.d. Θ_n:** Θ_{n+1} is independent of everything else, so (G_n) is Markov.
- **Markov Θ_n:** predicting Θ_{n+1} requires Θ_n, but G_n does not in general determine Θ_n. Hence (G_n) alone is *not* Markov; the augmented sequence $((G_n, \Theta_n))_n$ is.
- **HMM Θ_n:** same issue (a hidden state is needed); (G_n) is not Markov.

4.2 Distribution of $F_n(x)$ vs $G_n(x)$

Because $\Theta_1, \dots, \Theta_n$ are i.i.d., the joint distribution of any permutation of them is the same. In particular reversing the order leaves the joint law unchanged, so for each fixed n

$$F_n(x) \overset{d}{=} G_n(x).$$

The two *sequences* are however distributionally different. In (G_n), each new Θ_{n+1} is post-composed on the outside, so (G_n) is Markov (part 4.1). In (F_n), the new Θ_{n+1} is composed on the *inside*: $F_{n+1}(x) = F_n(f_{\Theta_{n+1}}(x))$, which depends on the entire history $\Theta_1, \dots, \Theta_n$ baked into F_n *plus* the new parameter at a different position. Hence (F_n) is not Markov in general, and $(F_1, F_2, \dots) \overset{d}{\neq} (G_1, G_2, \dots)$ as sequences.

4.3 Distribution of $F_{N+1}(x)$

By assumption $F_N(x) = X$ for every $x \in \mathbf{K}$, with $X \sim \pi$. Therefore

$$F_{N+1}(x) = F_N\big(f_{\Theta_{N+1}}(x)\big) = X,$$

because F_N is the constant map $\equiv X$. Hence $F_{N+1}(x) \sim \pi$.

4.4 Algorithm to Draw a Single Sample from π

```
Input: any x in K; access to Theta_1, Theta_2, ...
n <- 0
F <- identity on K
while F is not a constant function:
    n <- n + 1
    F <- F o f_{Theta_n}     // post-compose: append the new map on the right
return F(x)                   // equals X ~ pi
```

The loop is exactly the construction of F_n. By assumption it terminates at some random time N, at which $F_N \equiv X \sim \pi$, so the returned value is a single exact draw from π.

4.5 Comparison to Metropolis–Hastings

The algorithm of Section 4.4 returns an *exact* sample from π (no burn-in, no asymptotic bias), whereas Metropolis–Hastings only converges to π asymptotically and any finite-time sample is biased. If a coupling-from-the-past construction with finite expected coalescence time is available, the Propp–Wilson algorithm is therefore preferable for drawing one exact sample. The trade-off is that constructing suitable f_θ is a strong requirement and the coalescence time may have heavy tails; Metropolis–Hastings is far more general.

4.6 Validity of the Staircase Block Sampler

Let $\Omega = \{(x,t) : 0 \le t \le p(x),\ a \le x \le b\}$ be the area under p, and let $|B_k|$ be the area of block k. The kernel acts as follows: given $X_{i-1} \sim p$,

$$T_i \mid X_{i-1} \sim \text{Uniform}[0, p(X_{i-1})] \implies (X_{i-1}, T_i) \sim \text{Uniform}(\Omega).$$

Now select the block k^* containing (X_{i-1}, T_i). Conditional on landing in block k^*, (X_{i-1}, T_i) is uniform on B_{k^*}, and resampling (X_i, Y_i) uniformly on the same block preserves uniformity on B_{k^*}. Marginalising over the (random) block, (X_i, Y_i) is again uniform on Ω. Discarding Y_i leaves X_i marginally distributed as the projection of Uniform(Ω) onto the x-axis, which is exactly $p(x)$.

Yes, this is a valid sampler: p is invariant under the kernel.

5 The Entropy Term in the Free Energy

5.1 Geometric Argument for $M = 2$

For data point $i = 1$, $\widetilde{\mathcal{F}}$ depends on the responsibilities (r_1^1, r_1^2) only through the linear function

$$g(r_1^1, r_1^2) = \sum_{m=1}^{2} r_1^m \log P(\mathbf{x}_1, s_1 = m \mid \boldsymbol{\theta}_m, \boldsymbol{\pi}).$$

The contour lines of g are straight lines, and the constraint surface $r_1^1 + r_1^2 = 1$ is a line segment between the two simplex vertices $(1,0)$ and $(0,1)$. A linear function on a line segment attains its maximum at one of the endpoints. The maximiser is the vertex $r_1^m = 1$ where

$$m = \underset{m \in \{1,2\}}{\arg\max} \log P(\mathbf{x}_1, s_1 = m \mid \boldsymbol{\theta}_m, \boldsymbol{\pi}),$$

i.e. the data point is hard-assigned to the most likely component.

5.2 The Correct Free Energy Lower-Bounds the Log-Likelihood

Starting from the model log-likelihood for a single data point,

$$\log P(\mathbf{x}_i \mid \boldsymbol{\theta}, \boldsymbol{\pi}) = \log \sum_{m=1}^{M} \pi_m P_m(\mathbf{x}_i; \boldsymbol{\theta}_m) = \log \sum_{m=1}^{M} P(\mathbf{x}_i, s_i = m \mid \boldsymbol{\theta}_m, \boldsymbol{\pi}).$$

Multiply and divide each term by the responsibility r_i^m and apply Jensen's inequality (with log concave and $\sum_m r_i^m = 1$):

$$\begin{aligned}
\log \sum_m P(\mathbf{x}_i, s_i = m \mid \boldsymbol{\theta}_m, \boldsymbol{\pi}) &= \log \sum_m r_i^m \, \frac{P(\mathbf{x}_i, s_i = m \mid \boldsymbol{\theta}_m, \boldsymbol{\pi})}{r_i^m} \\
&\geq \sum_m r_i^m \, \log \frac{P(\mathbf{x}_i, s_i = m \mid \boldsymbol{\theta}_m, \boldsymbol{\pi})}{r_i^m} \\
&= \sum_m r_i^m \log P(\mathbf{x}_i, s_i = m \mid \boldsymbol{\theta}_m, \boldsymbol{\pi}) - \sum_m r_i^m \log r_i^m.
\end{aligned}$$

Summing over i,

$$\sum_i \log P(\mathbf{x}_i \mid \boldsymbol{\theta}, \boldsymbol{\pi}) \geq \underbrace{\sum_i \sum_m r_i^m \log P(\mathbf{x}_i, s_i = m \mid \boldsymbol{\theta}_m, \boldsymbol{\pi})}_{\widetilde{\mathcal{F}}} + \underbrace{\Big(- \sum_i \sum_m r_i^m \log r_i^m \Big)}_{H[\{r_i^m\}]} \; = \; \mathcal{F}, \tag{12}$$

with equality iff $r_i^m = P(s_i = m \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\pi})$. So $\mathcal{F}$ lower-bounds the log-likelihood.

5.3 $\widetilde{\mathcal{F}}$ is also a Lower Bound

The entropy of any discrete distribution is non-negative: $H[\{r_i^m\}] \geq 0$. Combined with (12),

$$\widetilde{\mathcal{F}} = \mathcal{F} - H[\{r_i^m\}] \leq \mathcal{F} \leq \sum_i \log P(\mathbf{x}_i \mid \boldsymbol{\theta}, \boldsymbol{\pi}).$$

Hence $\widetilde{\mathcal{F}}$ is also a (looser) lower bound on the log-likelihood.

5.4 Optimal Responsibilities for $\mathcal{F}_\tau$, and the Limit $\tau \to 0$

For $\tau > 0$,

$$\mathcal{F}_\tau(\{r_i^m\}, \dots) = \sum_{i,m} r_i^m \log P(\mathbf{x}_i, s_i = m \mid \boldsymbol{\theta}, \boldsymbol{\pi}) \; - \; \tau \sum_{i,m} r_i^m \log r_i^m.$$

Lagrangian for each i (constraint $\sum_m r_i^m = 1$):

$$\frac{\partial}{\partial r_i^m} \Big[\mathcal{F}_\tau + \lambda_i \big(\textstyle\sum_{m'} r_i^{m'} - 1 \big) \Big] = \log P(\mathbf{x}_i, s_i = m \mid \boldsymbol{\theta}, \boldsymbol{\pi}) - \tau (\log r_i^m + 1) + \lambda_i = 0.$$

Solving for r_i^m and normalising:

$$r_i^m \; \propto \; P(\mathbf{x}_i, s_i = m \mid \boldsymbol{\theta}, \boldsymbol{\pi})^{1/\tau} \implies r_i^m = \frac{P(\mathbf{x}_i, s_i = m \mid \boldsymbol{\theta}, \boldsymbol{\pi})^{1/\tau}}{\sum_{m'} P(\mathbf{x}_i, s_i = m' \mid \boldsymbol{\theta}, \boldsymbol{\pi})^{1/\tau}}. \tag{13}$$

This is a softmax with temperature τ. As $\tau \to 0^+$, $1/\tau \to \infty$ and the softmax becomes a hard arg max: $r_i^m \to 1$ for the largest $P(\mathbf{x}_i, s_i = m \mid \cdot)$ and 0 otherwise — exactly the geometric conclusion of Section 5.1.

5.5 Algorithm Recovered for an Isotropic Gaussian Mixture

With $\pi_m = 1/M$, $P_m(\mathbf{x}; \mu_m) = \mathcal{N}(\mathbf{x} \mid \mu_m, \mathbf{I})$, and the hard responsibilities of Section 5.1, the algorithm that maximises $\widetilde{\mathcal{F}}$ alternates: (i) assign each point to the cluster with nearest mean, and (ii) update means as cluster averages. This is exactly K**-means clustering** (with $K = M$).

6 HMM Posteriors

6.1 Bayesian Filtering Recursion

Under the HMM conditional independences — $s_t \perp\!\!\!\perp \mathbf{x}_{1:t-1} \mid s_{t-1}$ (Markov hidden chain) and $\mathbf{x}_t \perp\!\!\!\perp \mathbf{x}_{1:t-1} \mid s_t$ (output independence given the current state) — the filtering recursion is

$$P(s_t = i \mid \mathbf{x}_{1:t}) = \frac{A_i(\mathbf{x}_t) \sum_j \Phi_{ji}\, P(s_{t-1} = j \mid \mathbf{x}_{1:t-1})}{P(\mathbf{x}_t \mid \mathbf{x}_{1:t-1})}, \tag{14}$$

with normalisation $P(\mathbf{x}_t \mid \mathbf{x}_{1:t-1}) = \sum_i A_i(\mathbf{x}_t) \sum_j \Phi_{ji} P(s_{t-1} = j \mid \mathbf{x}_{1:t-1})$.

The recursion uses both CI properties: the Markov property of the hidden chain reduces the time-update sum to involve only $P(s_{t-1} \mid \mathbf{x}_{1:t-1})$ (no longer history of states), while output independence allows the observation update to multiply by $A_i(\mathbf{x}_t)$ alone (no past observations).

6.2 MAP M-Step for the Transition Matrix Φ

Define the two-slice posterior

$$\xi_t(j, i) = P(s_{t-1} = j,\, s_t = i \mid \mathbf{x}_{1:T}, \Phi^{\text{old}}, \boldsymbol{\pi}, \{A_k\}).$$

The terms of the EM free energy that depend on Φ, plus the Dirichlet log-prior on each row $\Phi_{i,\cdot} \sim \text{Dirichlet}(\boldsymbol{\lambda})$, are

$$\mathcal{F}(\Phi) = \sum_{t=2}^{T} \sum_{i,j} \xi_t(j, i) \log \Phi_{ji} + \sum_{i,j} (\lambda_j - 1) \log \Phi_{ij} + \text{const}.$$

Maximise under the row-stochasticity constraint $\sum_i \Phi_{ji} = 1$ for each j. Lagrangian (with multipliers μ_j):

$$L(\Phi, \boldsymbol{\mu}) = \sum_{j,i} \left[\sum_{t=2}^{T} \xi_t(j, i) + \lambda_i - 1 \right] \log \Phi_{ji} + \sum_j \mu_j \left(\sum_i \Phi_{ji} - 1 \right).$$

Setting $\partial L / \partial \Phi_{ji} = 0$:

$$\Phi_{ji} = \frac{\sum_{t=2}^{T} \xi_t(j, i) + \lambda_i - 1}{-\mu_j}.$$

Imposing $\sum_i \Phi_{ji} = 1$ fixes $-\mu_j$ and yields the MAP update:

$$\boxed{\hat{\Phi}_{ji} \;=\; \frac{\sum_{t=2}^{T} \xi_t(j, i) + \lambda_i - 1}{\sum_{i'} \Big(\sum_{t=2}^{T} \xi_t(j, i') + \lambda_{i'} - 1 \Big)}.} \tag{15}$$

This is the standard EM update for HMM transitions augmented by Dirichlet pseudo-counts $\lambda_i - 1$. The two-slice marginal can be obtained from forward-backward messages as $\xi_t(j, i) \propto \alpha_{t-1}(j)\, \Phi_{ji}^{\text{old}}\, A_i(\mathbf{x}_t)\, \beta_t(i)$.

6.3 Closed-Form Likelihood as a Matrix-Vector Product

Let $\boldsymbol{\alpha}_t \in \mathbb{R}^K$ have components $\alpha_t(i) = P(\mathbf{x}_{1:t}, s_t = i \mid \Phi, \boldsymbol{\pi}, \{A_k\})$. The forward recursion is

$$\alpha_t(i) = A_i(\mathbf{x}_t) \sum_j \alpha_{t-1}(j)\, \Phi_{ji}, \qquad \alpha_1(i) = \pi_i\, A_i(\mathbf{x}_1).$$

With the diagonal observation matrix $A_t = \mathrm{diag}(A_1(\mathbf{x}_t), \dots, A_K(\mathbf{x}_t))$ and noting $\sum_j \alpha_{t-1}(j)\Phi_{ji} = (\Phi^\top \boldsymbol{\alpha}_{t-1})_i$, the recursion in matrix form is

$$\boldsymbol{\alpha}_t = A_t\, \Phi^\top\, \boldsymbol{\alpha}_{t-1}, \qquad \boldsymbol{\alpha}_1 = A_1 \boldsymbol{\pi}.$$

Iterating from $t = 2$ to T,

$$\boldsymbol{\alpha}_T = A_T \Phi^\top A_{T-1} \Phi^\top \cdots A_2 \Phi^\top A_1 \boldsymbol{\pi}.$$

The likelihood is $P(\mathbf{x}_{1:T} \mid \Phi, \boldsymbol{\pi}, \{A_k\}) = \sum_i \alpha_T(i) = \mathbf{1}^\top \boldsymbol{\alpha}_T$, i.e.

$$\boxed{P(\mathbf{x}_{1:T} \mid \Phi, \boldsymbol{\pi}, \{A_k\}) \;=\; \mathbf{1}^\top\, A_T\, \Phi^\top\, A_{T-1}\, \Phi^\top \cdots A_2\, \Phi^\top\, A_1\, \boldsymbol{\pi}.} \tag{16}$$

6.4 Posterior on Φ and Direct Maximisation

With $\boldsymbol{\pi}$ and $\{A_k\}$ known and a Dirichlet prior $p(\Phi) = \prod_j \mathrm{Dirichlet}(\Phi_{j,\cdot} \mid \boldsymbol{\lambda})$, by Bayes' rule and (16),

$$p(\Phi \mid \mathbf{x}_{1:T}) \;\propto\; p(\Phi) \left(\mathbf{1}^\top A_T \Phi^\top A_{T-1} \Phi^\top \cdots A_2 \Phi^\top A_1 \boldsymbol{\pi} \right). \tag{17}$$

Direct maximisation is generally infeasible. The likelihood factor in (17) is a polynomial of degree $T-1$ in the entries of Φ (one $\Phi^\top$-factor per time step), and equivalently a sum of K^T products $\pi_{j_0} \prod_t \Phi_{j_{t-1} j_t} \prod_t A_{j_t}(\mathbf{x}_t)$ over hidden state sequences. Its log is the log of a sum-of-products, which is non-concave in Φ and has no closed-form stationary point. This is precisely why EM is used: each M-step replaces the intractable log-marginal-likelihood by the tractable expected complete-data log-likelihood whose maximiser is closed-form, namely (15).